PENETRATION TESTING REPORT

FOOTPRINTING AND NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Report detail	Value
Pentester Name (Cybersecurity Professional)	Emmanuel Eyo
Program/Batch	B084D-Networkwalks
Modules Completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Permission secured from client?	Yes
Evidence dates	10–12 November 2026
Client/Target	Public footprinting of networkwalks.com and host discovery on the trainee's Wi-Fi LAN, 192.168.8.0/24
Phrases Covered	**Phase 1** : Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

This report records the commands and observations contained in the Emmanuel's Week 2 evidence. The work shown is limited to public-information footprinting of networkwalks.com and a ping scan of the Emmanuel's local Wi-Fi subnet. The evidence does not show exploitation, credential testing, port enumeration, or vulnerability validation. The supplied files do not include the written authorization or rules of engagement; attach the internship task brief if it is required to document the approved scope.

Tool fingerprints and DNS responses are time-bound observations. They do not, by themselves, establish that a system is vulnerable or that a control is effective.

2. Introduction

This report summarizes two Week 2 cybersecurity exercises: collecting public domain and web-technology information for networkwalks.com, and discovering live devices on the trainee's own Wi-Fi LAN with Zenmap. The footprinting outputs were captured on 10 November 2026. The Windows network configuration and Zenmap host scan were captured on 12 November 2026.

The results below distinguish what the tools reported from what can be concluded. No exploit attempt or vulnerability confirmation is documented in the supplied evidence.

3. Tools Used

Tool	Environment	Purpose	Command / profile
WHOIS	Kali Linux	Retrieve public domain registration data.	whois networkwalks.com
WhatWeb	Kali Linux	Fingerprint publicly visible web technologies and HTTP responses.	whatweb networkwalks.com
Nslookup	Kali Linux	Resolve the domain through the configured DNS resolver.	nslookup networkwalks.com
Curl	Kali Linux	Inspect HTTPS response headers.	curl -I https://networkwalks.com
WAFW00F	Kali Linux	Identify a possible web application firewall.	wafw00f networkwalks.com
DNSRecon	Kali Linux	Enumerate DNS records and service information.	dnsrecon -d networkwalks.com
ipconfig	Windows	Read the active Wi-Fi IPv4 configuration and adapter MAC.	ipconfig
Zenmap / Nmap	Windows	Discover responding hosts in the local /24 subnet.	Ping Scan: nmap -sn 192.168.8.100/24

4. Activities Performed

4.1 Footprinting and Reconnaissance

The domain footprinting exercise used six Kali Linux tools against networkwalks.com. The first WHOIS attempt failed with a temporary name-resolution error for whois.verisign-grs.com. The evidence then shows successful WHOIS output. WhatWeb also showed initial name-resolution errors before a later run returned results.

WHOIS

The successful WHOIS responses identify GoDaddy as registrar, a creation date of 6 November 2019, and a registry expiry date of 6 November 2027. The registry response lists ns6135.hostgator.com and ns6136.hostgator.com; the registrar response lists those plus ns29.domaincontrol.com and ns30.domaincontrol.com. The registrant is listed through Domains By Proxy, and the registry response reports DNSSEC as unsigned. Because the two responses show different name-server sets, confirm the current delegation with a direct DNS NS query before treating either set as authoritative.

WhatWeb and Nslookup

WhatWeb reported an HTTP 301 redirect to HTTPS and HTTPS 200 responses for networkwalks.com, resolving to 192.232.216.135. Its output identified Apache, WordPress 7.1.4, WordPress Download Manager 3.3.76, jQuery 3.7.1, and Bootstrap 7.1.4. These are tool fingerprints and require confirmation against the owner's asset inventory before being treated as verified software versions. Nslookup, using resolver 8.8.8.8, returned 192.232.216.135.

Curl and WAFW00F

The curl -I request to https://networkwalks.com returned HTTP/2 200. The captured headers identify Apache, a WordPress REST API link at /wp-json/, and a secure, HttpOnly cookie. The response also includes a referrer-

policy header. These are observations of the public response; the presence of a REST API link or server header is not, by itself, a vulnerability. WAFW00F reported ModSecurity (SpiderLabs) after two requests. This is a tool identification result and does not demonstrate how well the WAF blocks attacks.

DNSRecon

DNSRecon reported "No answer" for its DNSSEC query, then found eight records. The output includes an SOA and NS record for ns6135.hostgator.com (50.87.144.87), an A record for networkwalks.com (192.232.216.135), an MX record for mail.networkwalks.com (192.232.216.135), TXT records including an SPF policy ending in ~all, several autodiscover SRV records, and a BIND version string of 9.16.23-RH. The DNSSEC message and software banner should be independently verified; neither result alone confirms a configuration weakness or exploitable issue.

4.2 Network Scanning with Zenmap

The Windows Wi-Fi adapter had IPv4 address 192.168.8.100 and subnet mask 255.255.255.0, which corresponds to LAN subnet 192.168.8.0/24. The default gateway and DNS server shown by ipconfig were 192.168.8.1. Zenmap ran the Ping Scan command nmap -sn 192.168.8.100/24 and reported 256 addresses scanned with two hosts up.

Live host	Identification from evidence	MAC address	Evidence source
192.168.8.1	Default gateway	3C:67:C9:0B:AA:05	Zenmap scan output
192.168.8.100	Emmanuel's Wi-Fi adapter / local PC	84:5C:F3:CF:A0:67	ipconfig Wi-Fi Physical Address; Nmap did not display this host's MAC

The scan identified two responding hosts. The supplied scan output displays a MAC address for 192.168.8.1; the local PC's MAC address is read from ipconfig, not from the Nmap output. The topology screenshot shows the same two-host network view.

5. Risk Analysis / Impact

Observation	Potential impact	Rating
Web and platform fingerprints were returned by WhatWeb.	Technology details can help prioritize later research, but the scan does not verify versions or identify a vulnerability.	Informational
DNS records and a DNS software banner were returned by DNSRecon.	Public records reveal hosting and mail-service relationships. A banner is not proof of an exposed or vulnerable DNS service.	Informational
Curl exposed the web server header and WordPress REST API links; WAFW00F identified ModSecurity.	These results describe visible services and a likely protective control. No weakness or WAF bypass was tested.	Informational
Zenmap found two live devices on the local /24 network.	Discovery supports asset inventory. The evidence does not establish whether either device is unauthorized.	Informational

No confirmed vulnerability was established by these exercises. The work consisted of information gathering and ping-based host discovery; it did not include port scanning, exploitation, or validation of the software versions reported by fingerprinting tools.

6. Recommendations

- **Confirm** the active authoritative name servers and DNSSEC status using direct DNS queries and the organization's DNS records.
- **Review** public server and platform fingerprints, and verify the reported WordPress, plugin, jQuery, Bootstrap, and BIND versions against the maintained asset inventory before deciding whether updates are needed.
- **Review** HTTP response headers and exposed API routes against the organization's intended configuration; avoid treating a publicly reachable REST endpoint as a vulnerability without validation.
- **Keep** the WAF enabled and monitored, and verify its policy through authorized testing rather than relying on product identification alone.
- **Compare** the two discovered LAN hosts with the approved device inventory and investigate any device that is not recognized.
- **Retain** the internship assignment or written scope with the report so the authorized target and allowed techniques are clear.

7. Conclusion

The Week 2 exercises documented public domain and web-response information for networkwalks.com and discovered two live hosts on the trainee's 192.168.8.0/24 Wi-Fi LAN. The observations include a successful DNS resolution to 192.232.216.135, tool-reported web technologies and ModSecurity, DNS record details, and a local host list with MAC information from the available evidence.

The work demonstrates basic reconnaissance and host discovery. The results are not a vulnerability assessment: software fingerprints, DNS responses, HTTP headers, and live-host responses require context and independent validation before security conclusions are drawn.

8. Evidence Collected

The figures below are reproduced from the trainee's evidence document and the screenshots shared for this report. DNSRecon results were supplied as text in the evidence document; no separate DNSRecon screenshot was embedded there.

Footprinting evidence

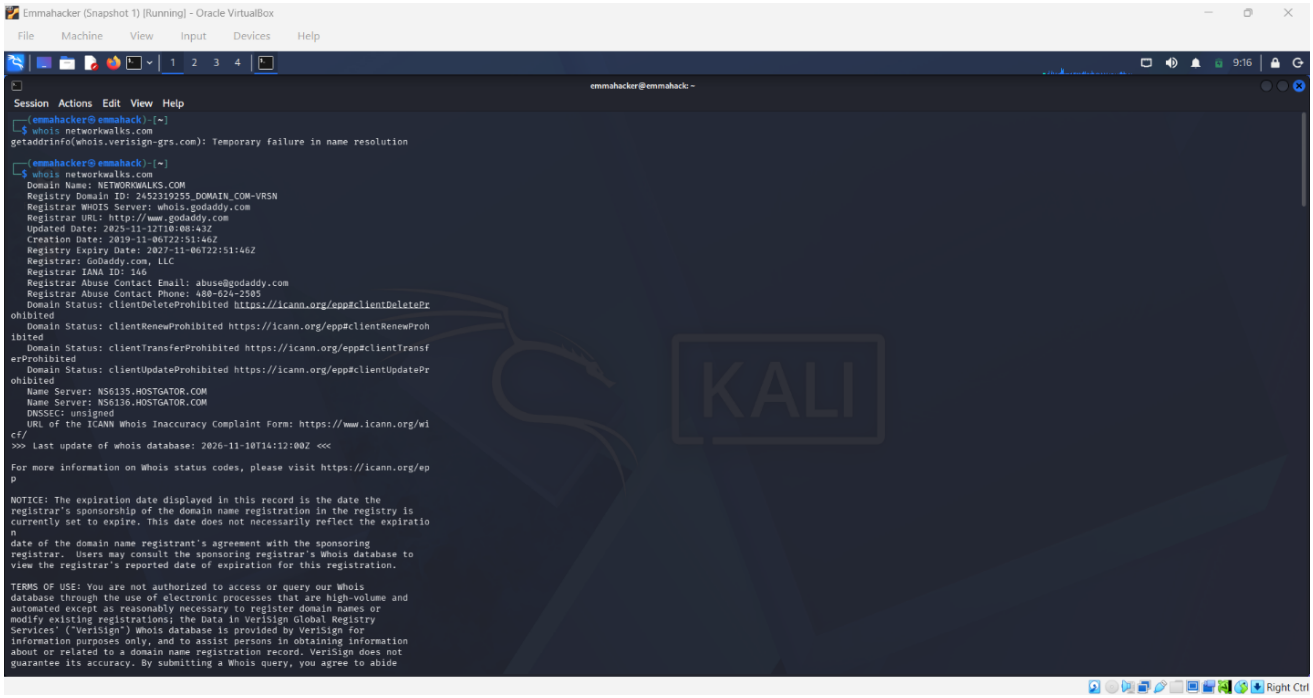


Figure 1. WHOIS first reports a temporary DNS resolution failure, followed by a successful query.

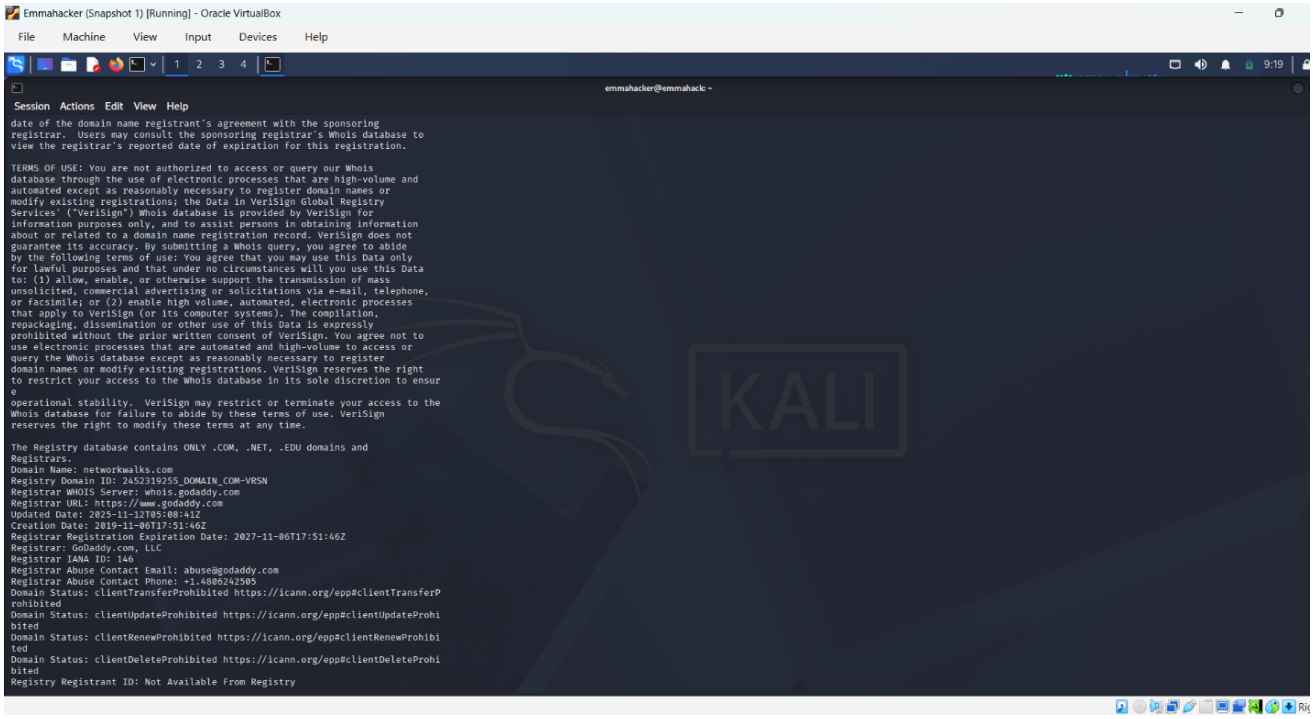


Figure 2. Continuation of the successful WHOIS response.

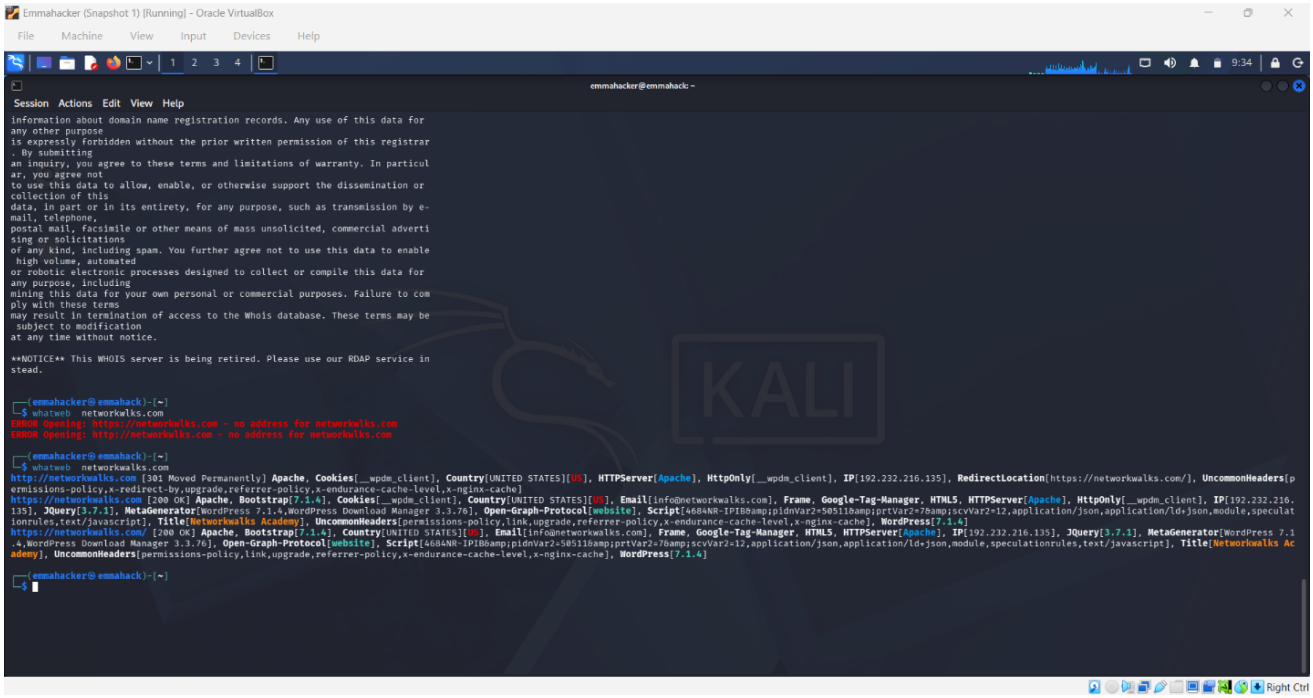


Figure 3. WHOIS output continuation and WhatWeb retry/results.

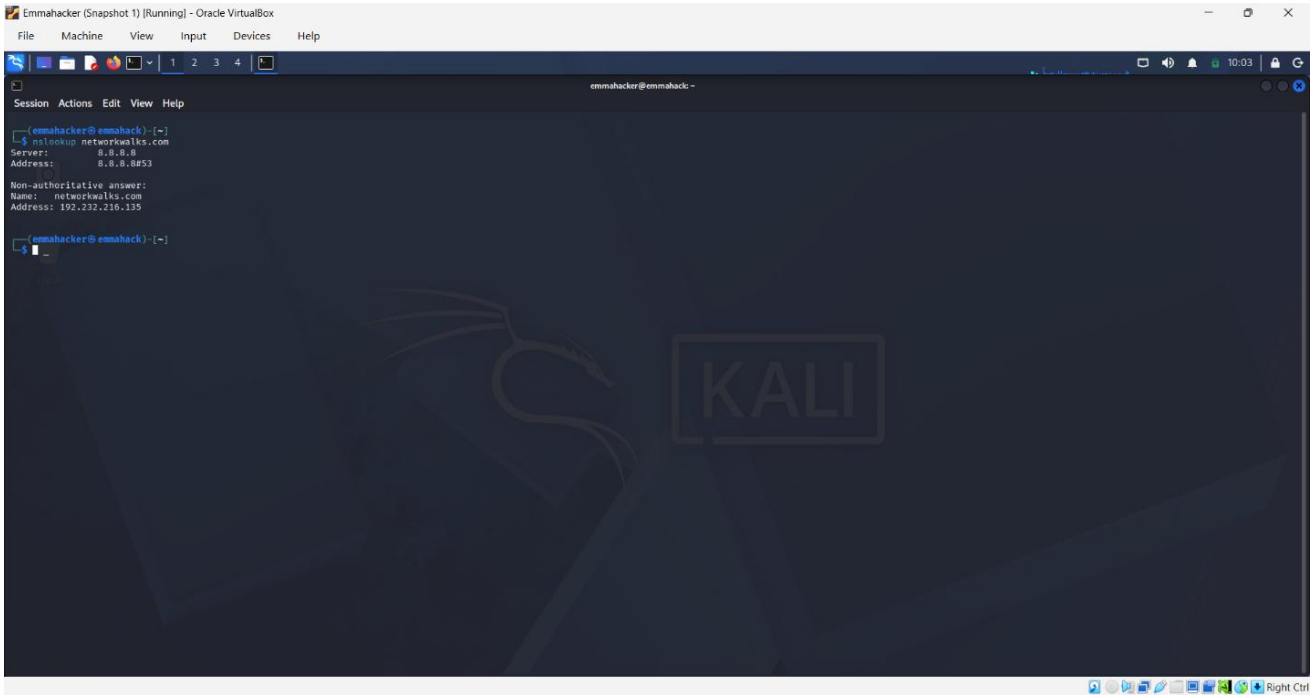


Figure 4. Nslookup resolves networkwalks.com to 192.232.216.135.

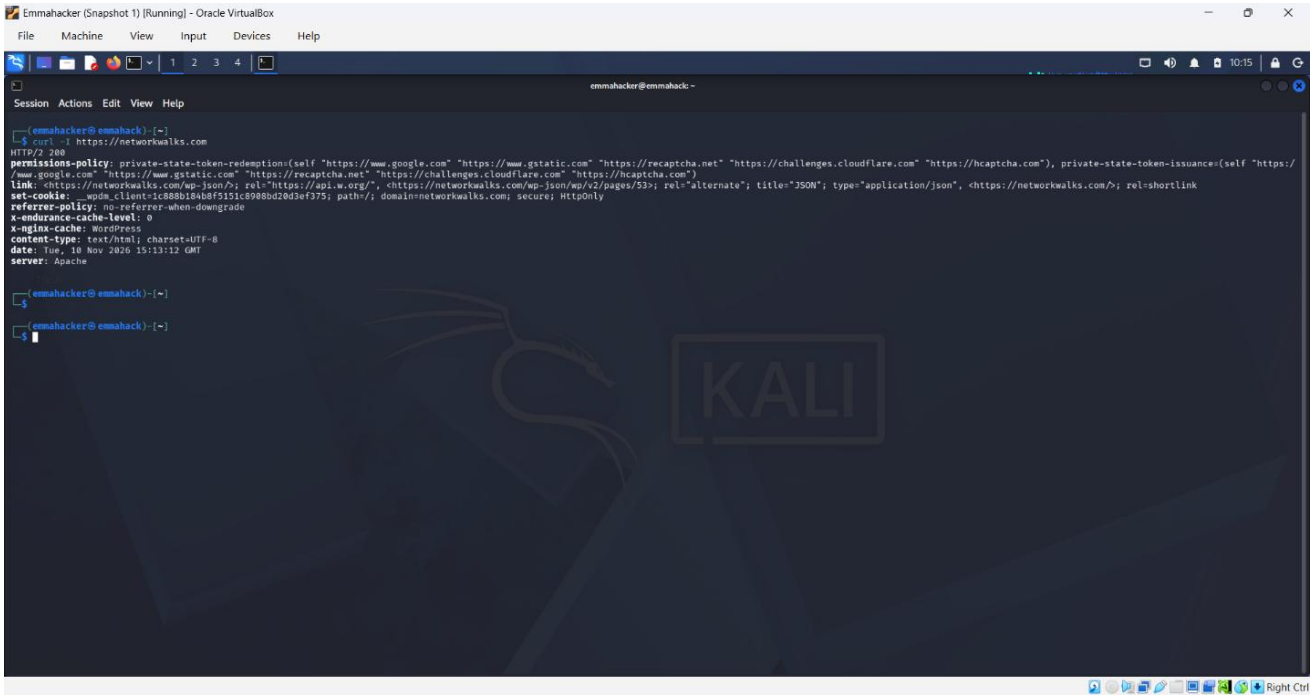


Figure 5. Curl response headers from the HTTPS endpoint.

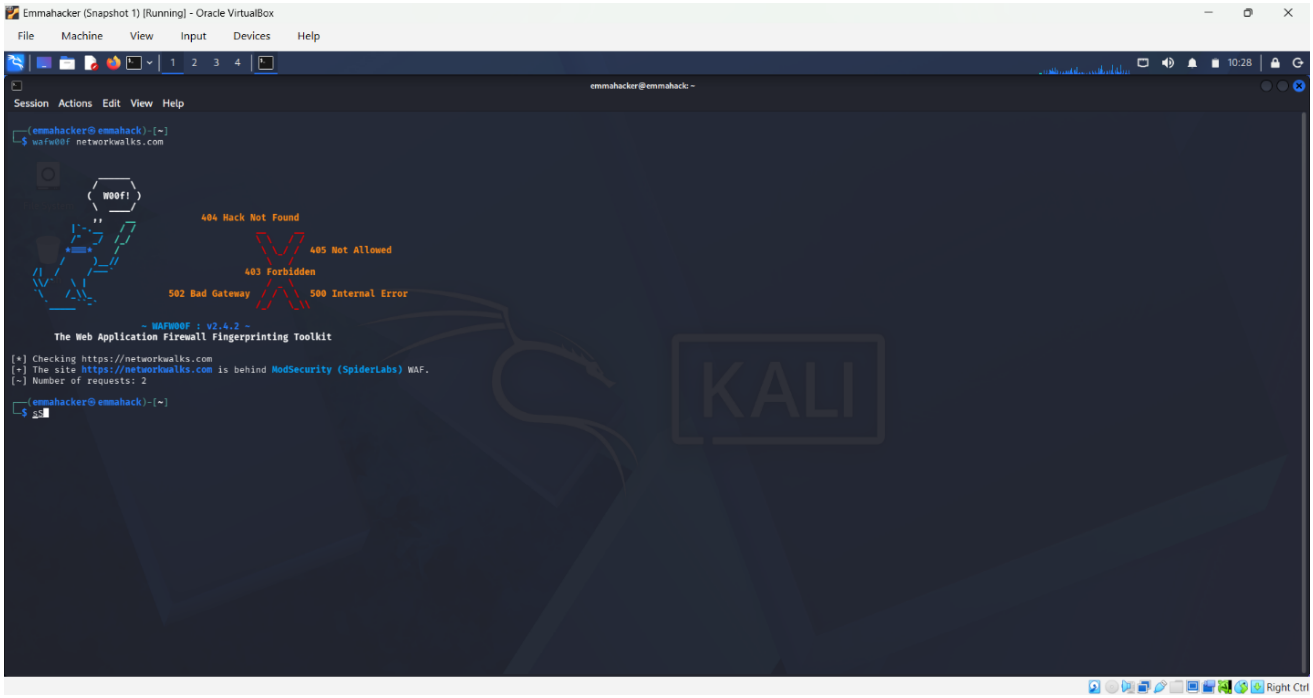


Figure 6. WAFW00F reports ModSecurity (SpiderLabs).

Local network evidence

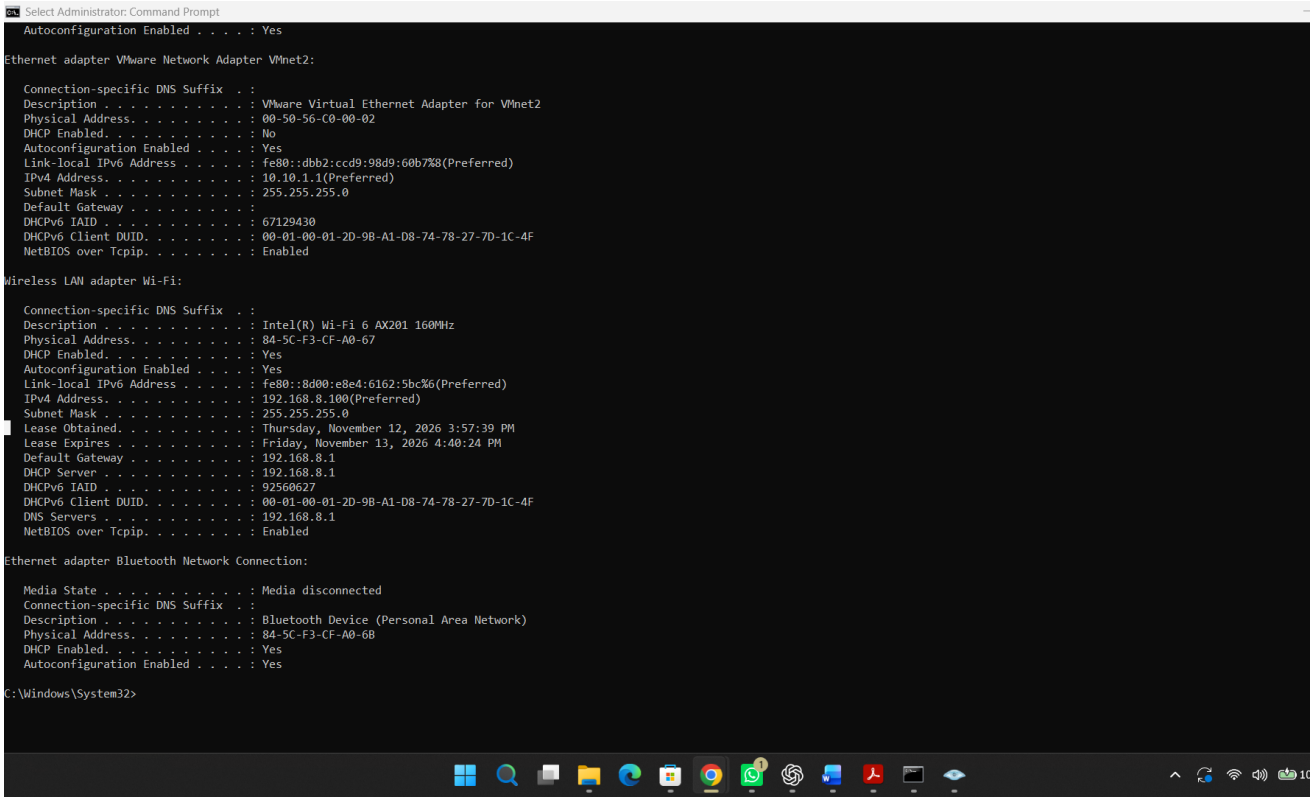


Figure 7. Windows ipconfig output showing the Wi-Fi IPv4 address, subnet mask, gateway, and adapter MAC.

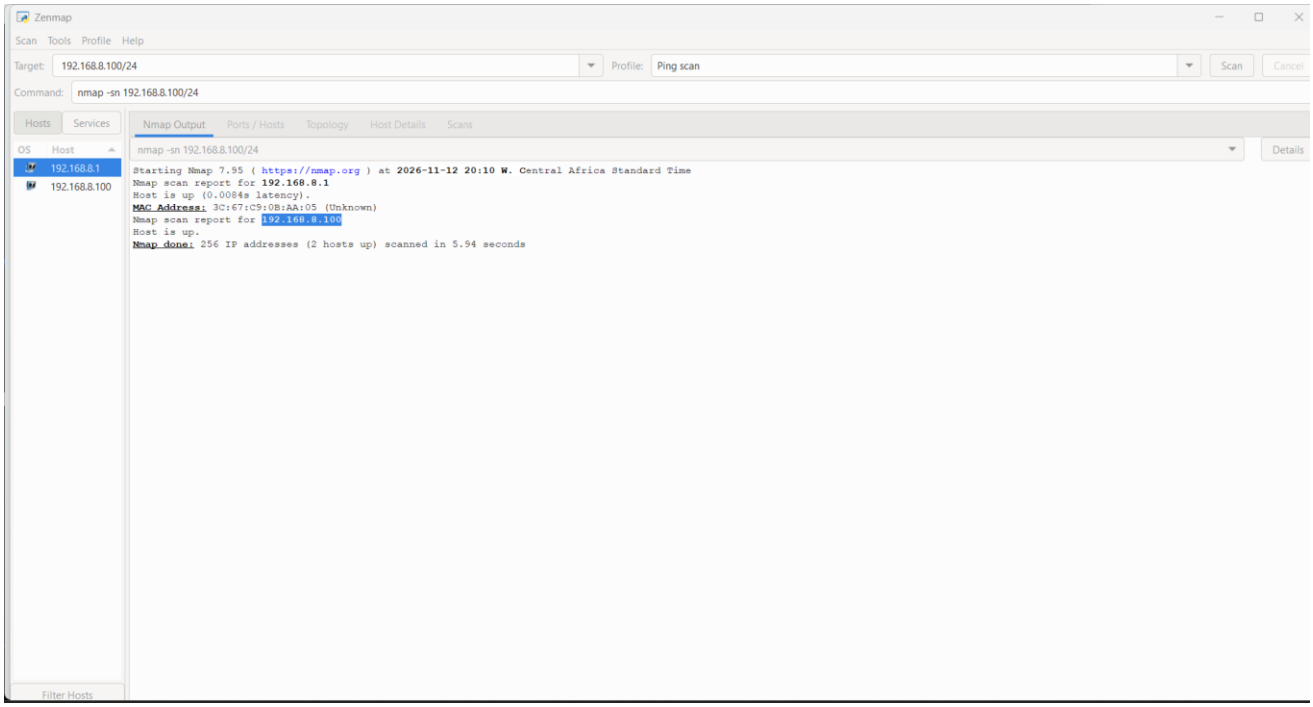


Figure 8. Zenmap Ping Scan output showing two live hosts on 192.168.8.0/24.

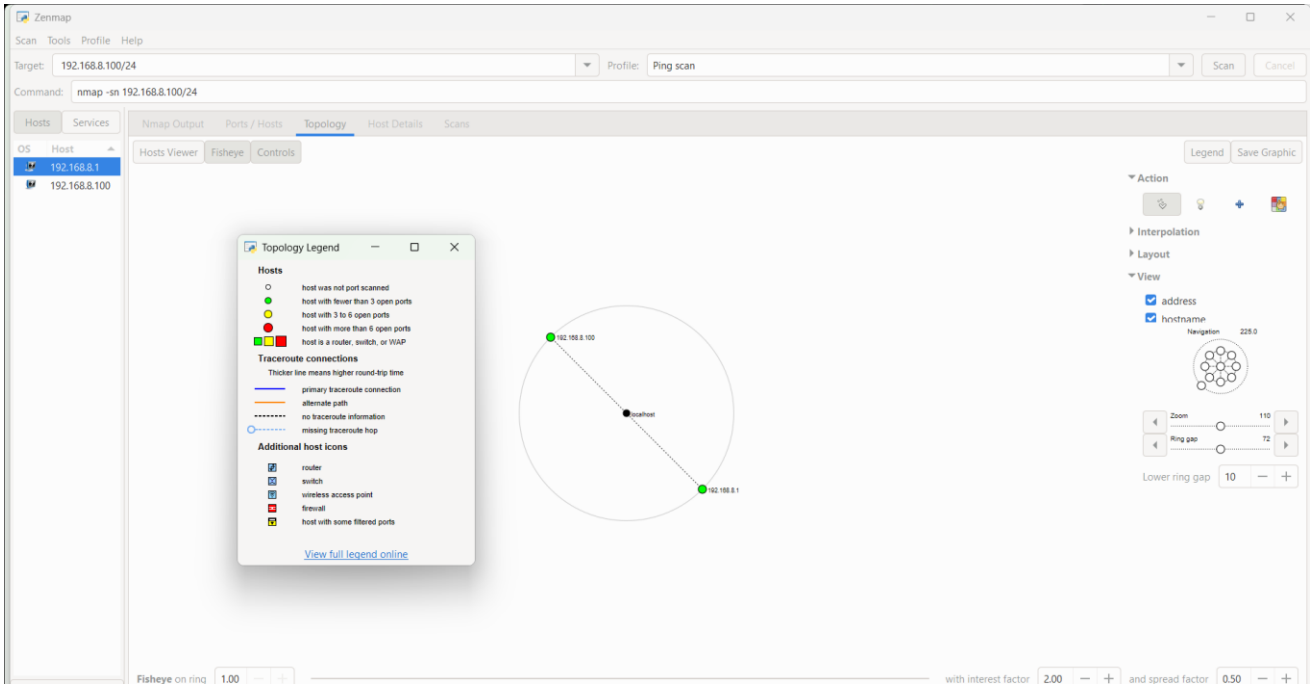


Figure 9. Zenmap topology view of the discovered local network.

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